MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE 8NW6

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:

First article required.

Construction

First - Copper conductor, uncoated, Navy Standard size 6(7).

Second - Separator, polyester type G of MIL-I-631, opaque white, applied either helically or longitudinally to give 100 percent coverage.

Third - Cross-linked thermosetting polyethylene insulation, colored black. Nominal thickness 0.030-inch. Standard Identification Code applied by method 1.

Fourth - Eight conductors cabled together with a lay not greater than 24 times the pitch diameter. Either non-fibrous or a combination of non-fibrous and fibrous fillers shall be used to form a firm, well-rounded assembly.

Fifth - Binder tape of 0.002 inch thick, corrugated polyester, type G of MIL-I-631.

Sixth - Standard thermoplastic jacket (gray). Nominal thickness 0.040-inch. Maximum overall diameter 0.670-inch. Cable surface marking required.

EXAMINATION AND TESTS:

	Requirements:
Basic electrical:	
Conductor resistance - At 25° Celsius (C) ohms/1000 feet, maximum	1.72
Voltage withstand - Condr. to condr., volts, root mean square (rms), minimum	4000
Insulation resistance - Megohms/1000 feet, minimum	4200
Group A:	
Visual and dimensional.	
Group B:	
Cold bending, cable - 5-inch mandrel diameter, maximum	No damage
Physicals (unaged)	
Insulation	
Tensile strength - pounds per square inch gage (lb/in^2g), minimum	1800
Elongation - percent, minimum	250
Jacket	
Tensile strength - lb/in^2g, minimum	1800
Elongation - percent, minimum	225
Group C:	
Permanence of printing, (conductor) - cycles, minimum	25
Permanence of printing, (jacket) - cycles, minimum	250
Cable filler removability	
Flammability - inches, maximum	2
Heat distortion - percent of unaged, maximum	30

FSC 6145

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	Requirements:
Drip - At 95° ± 1°C.	Zero
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum.	75
Elongation - percent of unaged, minimum . . .	75
Jacket	
Hot oil, immersion	
Tensile strength - lb/in^2g, minimum	1400
Elongation - inches, minimum.	1

FIRST ARTICLE INSPECTION

First article inspection shall include basic electrical plus all of Groups A, B and C plus the following:

Cold working (minus 20° ± 2°C)	.No damage
Cable aging and compatibility (125° ± 5°C)	.No failure

UNIT ORDERING LENGTH: 2000 feet (nominal)

Preparing activity:
Navy - SH
(Project 6145-N256)

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